WEAKLY ASSOCIATED POINTS UNDER A FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a reduction to algebra: the affine translation is done in place and the algebra content is delegated to the cited Algebra lemma; that delegation is recorded here for the reader.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Weakly associated points of a sheaf on an affine scheme are the weakly associated primes. Lemma lemma-weakly-associated-affine-open, used at proof lines 886-891. Translates both sides into algebra.
- Weakly associated primes under a finite ring map. Algebra, Lemma algebra-lemma-weakly-ass-finite-ring-map, used at proof lines 891-894. Is the algebra statement that finishes the proof.

Source reference. Original lemma 05E1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6974–6974. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 056M: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6715–6715. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a finite morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $\mathit{WeakAss}(f_*\mathcal{F}) = f(\mathit{WeakAss}(\mathcal{F}))$.*

Proof. We may assume X and S affine, so $X \to S$ comes from a finite ring map $A \to B$. Write $\mathcal{F} = \widetilde{M}$ for some B-module M. By Lemma 056M the weakly associated points of $\mathcal{F}$ correspond exactly to the weakly associated primes of M. Similarly, the weakly associated points of $f_*\mathcal{F}$ correspond exactly to the weakly associated primes of M as an A-module. Hence the lemma follows from Algebra, Lemma 05E1. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.